Structural Consequences of Observational Incompleteness

From Chemistry to Evolution

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Abstract

The Observational Incompleteness (OI) framework develops a conditional lattice/gauge construction [1, 2]. Within A4's assumed periodic-lattice class, E5–E7 select $d = 3$ subject to the continuum conditions of [2, §3.1–§3.2]; identifying the geometric link space with the physical carrier requires H-link, with H-spin' and H-χ' entering the generation and chirality readings. This paper explores consequences for orbital structure, the periodic table, bonding geometry, scale hierarchies and chemistry, conditional on the stated spin, statistics, atomic and matter assumptions. Three generations permit a physical CKM phase, whose value is a solution-specific input; a mass hierarchy also requires the stated Yukawa and taste-breaking assumptions. Chiral SU(2), conditional on H-χ' [2, §4.8], supplies a parity-violating interaction whose molecular energy difference and preferred sign require the stated molecular and observed weak-sector inputs. The viable fraction of the chosen 18-dimensional parameter space is estimated at ~16%.

Counting every possible peptide of length 50 ($\sim 20^{50}$), Kauffman's random-catalysis model gives a nominal margin $Np \sim 10^{56}$ over its autocatalytic threshold; the model's N counts co-present species, so the margin holds only to the extent that such diversity is available. Template replication requires three independently structural capabilities (linear polymers, complementary pairing, catalysis). A template replicator can realize C1–C4, its sequence the persistent record that copying reads back; C1–C4 do not imply replication, growth or catalytic closure. RNA's dual role as template and catalyst is empirical support for the RNA-world hypothesis. Given a replicator with sufficient copying fidelity in a feedstock-rich environment, population dynamics gives exponential growth while resources last, and Darwinian evolution follows from imperfect template replication. The chain has a natural terminus at biological evolution; the continuation from evolution through general intelligence, AI, the BQP characterization theorem for embedded observers, and the framework's position on the P vs NP question is the subject of the companion paper *Embedded Computation in the Incompleteness of Observation Framework* [38].

The same C1–C4 conditions that diagnose an accessible non-Markovian real-

ization cosmologically — finite-horizon memory in the visible law, not generic quantum non-Markovianity — are satisfied in proteins ($\tau_S/\tau_B \sim 10^{-9}$ to 10^{-12}), predicting non-Markovian enzyme kinetics consistent with single-molecule experiments. The ratio τ_S/τ_B sets the bath's retention window, of order τ_B/τ_S events, not the size of the correction; in quantum computing (10^{-3}) and quantum sensing (10^{-6}) that window falls within device run lengths, where memory-bearing noise is predicted to be engineering-relevant. The framework identifies a new design space — engineered C1–C4 architectures — in which the environment is programmable quantum memory rather than noise.

Status — conjectural extension. This paper applies the core framework to quantum computation and complexity, and is offered as a *conjecture*, not an established result: the central claim — that the framework's emergent dynamics realizes the complexity class BQP — is a theoretical conjecture rather than a proven correspondence; it would be supported by a rigorous derivation from the finite substratum to BQP and undercut if that dynamics proves efficiently classically simulable or fails to reproduce known quantum-complexity separations. It carries no evidential weight for the core framework — which stands or falls independently on its physics — and should be read as a structural possibility to be tested, not as support for the framework.

1. Introduction

The fine-tuning problem in physics rests on a counterfactual: if the laws of physics or the values of fundamental constants were different, the universe would not permit complex structures, and we would not exist to observe it. The anthropic principle elevates this observation into an explanatory principle — our existence selects the laws from an ensemble of possibilities.

The OI framework [1, 2] challenges the premise. The companion papers prove that an embedded observer in a finite deterministic system with conditions C1–C4 (coupling, record persistence, sufficient capacity, history readback) has a visible-sector description with positive conditional memory at an accessible order, and that this description, like every finite observable law, admits an exact fixed-basis unitary representation [1], and that, under the hypotheses stated there, the dynamics whose form the framework's additional structural requirements fix up to one coefficient determines the Standard Model's complete structure [2]. If the laws are derived rather than contingent, the largest source of anthropic variation — the structure of the laws themselves — is eliminated.

This paper traces the consequences. §2 establishes the structural chain from (S, φ) to the full preconditions for organic chemistry — orbital structure, the periodic table, carbon's bonding geometry, the nuclear-atomic scale hierarchy given the coupling values and masses, water's solvent properties, and the con-

ditions for a thermal window for dynamic chemistry. §3 estimates the viable fraction of the parameter space using existing literature. §4 traces the chirality chain from the electroweak sector to biological homochirality, its first link conditional on the framework's open hypothesis H-χ' [2, §4.8]. §5 extends the chain through autocatalytic networks, template replication, the origin of life (heredity in a template replicator as molecular C1–C4 memory), and Darwinian evolution — closing with a forward-pointer to the companion paper [38] that continues the chain through general intelligence, AI, and the BQP characterization for embedded observers. §6 addresses the existence question. §7 applies C1–C4 to biological systems, predicting non-Markovian enzyme kinetics with implications for drug design. §8 identifies where OI corrections reach engineering relevance — quantum computing, quantum sensing, and precision metrology — and proposes three engineering frontiers: engineered partitions (the bath as programmable memory), non-Markovian quantum computation (P-indivisible gate sequences), and quantum materials with designed C1–C4 architectures.

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2. The Structural Chain

2.1 From (S, φ) to orbitals

Within A4's assumed periodic-lattice class, the physical filters E5–E7 select $d = 3$ subject to the continuum conditions [2, §3.1–§3.2]. Integer rank is an input of A4; renormalizability of the specified Yang–Mills coupling is a downstream consequence. In three spatial dimensions, the rotation group is SO(3), whose irreducible representations are labeled by angular momentum quantum number $l = 0, 1, 2, 3, ...$ with $(2l + 1)$ states each. These are the atomic orbitals: s ($l = 0$, spherical, 1 orientation), p ($l = 1$, dumbbell, 3 orientations), d ($l = 2$, 5 orientations), f ($l = 3$, 7 orientations).

On this selected three-dimensional continuum branch, rotational symmetry supplies the SO(3) orbital algebra. In $d = 2$, angular momentum has a single component — only circular harmonics exist, with no three-dimensional bonding geometry. In $d \geq 4$, the Coulomb potential $V(r) \sim r^{2-d}$ does not support stable bound states [3, 4]: classical orbits spiral into the nucleus, and the quantum Hamiltonian is unbounded below.

2.2 From orbitals to the periodic table

The framework derives spin-1/2 fermions from the staggered structure on the d = 3 lattice [2, §4.2]. The spin-statistics theorem — a consequence of the emergent QFT's Lorentz invariance [2, §3.1] — gives the Pauli exclusion principle: each orbital holds at most 2 electrons (spin up and spin down). The shell capacities follow:

$$s : 2, \quad p : 6, \quad d : 10, \quad f : 14$$

The periodic table's period lengths — 2, 8, 8, 18, 18, 32 — are determined by the filling order of these shells. This orbital counting is conditional on the selected $d = 3$ branch and the stated spin and statistics assumptions. The dimension selection inherits A4, E5–E7 and the continuum conditions [2, §3.1–§3.2].

2.3 Carbon's bonding geometry

Element 6 (carbon) has electron configuration $1s^2\ 2s^2\ 2p^2$ — four valence electrons. In d = 3, four valence electrons admit three hybridization modes:

sp^3 hybridization: Four equivalent orbitals pointing to the vertices of a regular tetrahedron (bond angle 109.5°). This is the geometry of methane (CH_4), diamond, and the backbone of all organic macromolecules — proteins, nucleic acids, lipids, carbohydrates.

sp^2 hybridization: Three planar orbitals at 120° plus one π bond. This is the geometry of graphite, benzene, and aromatic chemistry.

The *possibility* of these hybridizations — and hence the possibility of three-dimensional macromolecular architecture — is a structural consequence of the angular momentum algebra in d = 3.

2.4 Beyond carbon: the extended structural chain

Four additional structural results extend the chain beyond orbital chemistry.

Matter-antimatter asymmetry. The three-generation branch inherits H-link and H-spin' [2, §4.7] and permits a physical CKM phase; its actual value is a solution-specific input [2, §5.4]. A nonzero CP-violating phase supplies one of the Sakharov conditions for baryogenesis [5]. The weak-chirality reading of partition-induced parity violation additionally assumes H-χ' [2, §4.8, Theorem 13]. Whether these ingredients produce the observed asymmetry depends on the remaining baryogenesis dynamics and parameters.

Mass hierarchy. On the stated physical-carrier and matter branch, the Higgs mechanism [2, §4.7] gives fermion masses through Yukawa couplings. The taste-breaking construction permits generation-dependent couplings, but the existence of the observed light fermions also requires the stated mass and coupling inputs; it is not fixed by the Higgs field alone.

Nuclear binding. SU(3) color confinement [2, §4.6] produces bound hadrons. The existence of multi-nucleon bound states (nuclei) requires specific parameter values (§3), but the *mechanism* — color confinement binding quarks into hadrons, residual strong force binding hadrons into nuclei — is structural.

Chiral chemistry. Chiral SU(2) — conditional in the framework on the taste-selectivity hypothesis H-χ' [2, §4.8], and realized in the observed electroweak sector — produces parity violation in all weak processes. This propagates to molecular physics as a parity-violating energy difference between mirror-image molecules (§4).

2.5 The scale hierarchy

The framework derives two independent gauge groups — SU(3) and U(1) — with independent coupling strengths [2, §6]. Given their values and the masses, the two couplings set two widely separated energy scales:

$$E_{\text{nuclear}} \sim \Lambda_{\text{QCD}} \sim \text{MeV}, \qquad E_{\text{atomic}} \sim \alpha^2 m_e c^2 \sim \text{eV}$$

The ratio $E_{\text{nuclear}}/E_{\text{atomic}} \sim 10^6$ is parametric: the two independent gauge groups supply two independent couplings, and the separation of the resulting scales rests on the couplings' and masses' values. The universal induced coupling $1/\alpha_0 = 23.25$ at the matching scale $\mu = M_{\text{Pl}}$ is itself structural — determined by $N_f = 6$ and the Dynkin index (ambient coefficient; for hypercharge under H-observer-bundle and H-Y-vertex, [SM §6.5]) — and the framework reproduces all three SM gauge couplings at M_Z through non-perturbative gauge self-energy corrections and RG running, with three threshold parameters fitted to the three observed couplings [2, §6.3]. This separation has a consequence: nuclei are stable against chemical reactions. Chemistry rearranges electrons (eV) without touching nuclei (MeV). Atoms are permanent building blocks that can be assembled, disassembled, and reassembled without being destroyed.

2.6 Water

Element 8 (oxygen): configuration $1s^2\ 2s^2\ 2p^4$ — six valence electrons, two unpaired. The sp^3-like hybridization in d = 3, distorted by two lone pairs, gives a bond angle of ~104.5°. This is the geometry of water (H_2O).

Water's anomalous properties follow from this geometry. The bent structure creates a permanent dipole moment. The lone pairs enable hydrogen bonding — a secondary interaction (~0.2 eV) intermediate between covalent bonds (~3 eV) and thermal energy (~0.025 eV at 300K). Hydrogen bonding produces: high specific heat (thermal buffering for chemical reactions), density maximum near 4°C (ice floats, insulating liquid water below), and exceptional solvent capability (dissolving ionic and polar molecules for solution-phase chemistry).

The *possibility* of a molecule with these properties is structural: element 8's electron configuration in d = 3 with the derived orbital algebra guarantees a bent dihydride with lone pairs capable of hydrogen bonding. The *existence* of a liquid phase at specific temperatures is parametric.

2.7 The thermal window

For chemistry to support complexity, a thermal window must exist where bond energies (~1–5 eV) $\gg kT$ (bonds are stable), $kT > E_{\text{activation}}$ (reactions proceed), and a liquid solvent exists. The first condition, $E_{\text{bond}} \gg kT$, is a statement about the selected molecules at the selected temperature: the coupling

values and masses (§2.5) set the atomic scale of bond energies, but the nuclear–atomic scale separation does not itself establish the inequality. The second and third — activation energies low enough for reactions to proceed at such a temperature, and a solvent that is liquid there — are further conditions, set by the molecules available rather than by the scale separation, and the separation does not establish that the three overlap. Where they do, chemistry is *stable but dynamic*.

2.8 Summary of the structural chain

The dimensional premise in these chains is conditional on A4, E5–E7 and the continuum conditions [2, §3.1–§3.2]; the downstream steps retain the field-theory and atomic assumptions cited above.

$$(\mathrm{S},\,\varphi) \xrightarrow{\text{conditional}} d = 3 \xrightarrow{\mathrm{SO}(3)} \text{orbitals} \xrightarrow{\text{Pauli}} \text{periodic table} \xrightarrow{\text{element 6}} \text{tetrahedral carbon}$$

$$(\mathrm{S},\,\varphi) \xrightarrow{\text{conditional}} 3 \text{ generations} \xrightarrow{\text{CKM}} \text{CP violation} \xrightarrow{\text{Sakharov}} \text{baryogenesis possible}$$

$$(\mathrm{S},\,\varphi) \xrightarrow{\text{conditional}} \text{Higgs} \xrightarrow{\text{taste-breaking}} \text{mass hierarchy} \to \text{light fermions} \to \text{stable atoms}$$

$$(\mathrm{S},\,\varphi) \xrightarrow{\text{conditional}} \mathrm{SU}(3) \xrightarrow{\text{confinement}} \text{hadrons} \xrightarrow{\text{residual force}} \text{nuclei possible}$$

$$(\mathrm{S},\,\varphi) \xrightarrow{\text{conditional}} \mathrm{SU}(3) + \mathrm{U}(1) \text{ independent} \xrightarrow{\text{coupling values, } m_e} \text{scale hierarchy} \to \text{atoms are permanent buildin}$$

$$(\mathrm{S},\,\varphi) \xrightarrow{\text{conditional}} \text{element 8 in d} = 3 \to \text{bent } \mathrm{H_2O} \xrightarrow{\text{H-bonding}} \text{anomalous solvent}$$

$$(\mathrm{S},\,\varphi) \xrightarrow{\text{conditional; coupling values, } m_e} \text{scale hierarchy + orbital structure} \xrightarrow{\text{activation energies, solvent}} \text{thermal window}$$

$$(\mathrm{S},\,\varphi) \xrightarrow{\text{H-}\chi\text{'}} \text{chiral } \mathrm{SU}(2) \xrightarrow{\text{PVED}} \text{L-amino acid preference}$$

These chains combine results of [1, 2] with textbook relations on the selected three-dimensional continuum branch, conditional on A4's lattice class, E5–E7, the continuum conditions, and the stated spin, statistics and atomic assumptions. Each downstream link retains its own hypotheses; the chiral-SU(2) arrow additionally assumes H-χ' [2, §4.8]. The conclusions about atoms, nuclei, the periodic table, organic chemistry, matter-antimatter asymmetry and chirally selected molecules inherit these conditions.

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3. The Viable Parameter Space

3.1 The question

The structural chain establishes that organic chemistry is *possible*. For it to be *actual*, the 18 free parameters of the SM must fall in a viable region. How large is this region?

Traditional fine-tuning studies [6, 7, 8] vary both the structure and the parameters. Here the structure is held fixed on the selected lattice/gauge branch, reducing the question to parameters alone: the gauge group, dimension, generation count, and discrete symmetries are not varied (§3.4).

3.2 The most constrained parameters

Not all 18 parameters are equally constrained. The literature identifies four as critical for complex chemistry:

The fine-structure constant α. Governs atomic binding, molecular bond strengths, and the balance between electromagnetic and thermal energies. Stable atoms require $\alpha \lesssim 1$ (above this, relativistic effects destabilize inner shells). Complex chemistry requires bond energies $\gg kT$ at habitable temperatures, giving $\alpha \gtrsim 10^{-3}$ [8]. The viable range spans roughly two orders of magnitude: $10^{-3} \lesssim \alpha \lesssim 10^{-1}$, with our universe at $\alpha \approx 1/137 \approx 7.3 \times 10^{-3}$ — within the viable range but not near its center.

The electron-to-proton mass ratio m_e/m_p. Determines the separation of nuclear and atomic scales. Stable atoms with a rich periodic table require $m_e/m_p \ll 1$ (so electrons orbit far from the nucleus). Complex chemistry breaks down for $m_e/m_p \gtrsim 10^{-1}$ (electron clouds too compact for directional bonding). The viable range is roughly $10^{-5} \lesssim m_e/m_p \lesssim 10^{-1}$, spanning four orders of magnitude. Our value: $m_e/m_p \approx 5.4 \times 10^{-4}$.

The quark mass ratio $m_d - m_u$. Controls the proton-neutron mass difference and hence nuclear stability. If $m_d - m_u$ changes sign (proton heavier than neutron), hydrogen is unstable — catastrophic for chemistry. If $m_d - m_u$ is too large, no bound nuclei beyond hydrogen exist. The viable range is approximately $1 \lesssim (m_d - m_u)/\text{MeV} \lesssim 5$ [9], about a factor of 5. Our value: $(m_d - m_u) \approx 2.5$ MeV.

The strong coupling α_s. At the QCD scale (~1 GeV), α_s determines nuclear binding. If α_s is ~10% smaller, the deuteron is unbound and big bang nucleosynthesis fails to produce helium — but stellar nucleosynthesis could still produce carbon [8]. If α_s is ~30% larger, di-protons are bound, stars burn through hydrogen catastrophically fast. The viable range is roughly $0.8\alpha_s^{\text{obs}} \lesssim \alpha_s \lesssim 1.3\alpha_s^{\text{obs}}$, about a factor of 1.6 [10].

3.3 The viable fraction

The four critical parameters span roughly:

Parameter	Viable range (orders of magnitude)	Log-fraction
α	~2 orders within ~3 available	~60%
m_e/m_p	~4 orders within ~6 available	~65%
$m_d - m_u$	factor of ~5 within ~10 available	~50%
α_s	factor of ~1.6 within ~2 available	~80%

The joint viable fraction, treating the parameters as independent (a conservative assumption — correlations generally enlarge the viable region), is roughly:

$$f_{\text{viable}} \sim 0.6 \times 0.65 \times 0.5 \times 0.8 \approx 16\%$$

This is a rough estimate, but the order of magnitude is robust: the viable fraction is $\mathcal{O}(10\%)$, not $\mathcal{O}(10^{-50})$. The remaining 14 SM parameters (heavy quark masses, lepton masses, mixing angles) are less constrained — they affect the details of nuclear and atomic physics but not the existence of carbon chemistry.

3.4 Comparison to fine-tuning claims

Comparisons of fine-tuning depend on which structural choices and parameters are varied. The estimate here holds the selected three-dimensional lattice/gauge branch fixed and varies the chosen parameter subspace. It does not exclude other dimensions or gauge structures using embedded observation alone. The ~16% estimate applies only to the stated ranges and weighting assumptions.

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4. The Chirality Chain

4.1 From the partition to parity violation: the conditional first link

The OI partition supplies the grading; the chirality of the weak force enters conditionally. The trace-out over the hidden sector leaves an ε-graded visible sector on which spin- and taste-chirality coincide, with the kinetic operator coupling only across sublattices ($D_{ee} = 0$, $D_{oo} = 0$) [2, §4.8, Theorem 13].

Whether the weak coupling is chiral on that sector reduces to the taste-chirality selectivity of the SU(2) embedding — hypothesis H-χ', stated as open in [2, §4.8]. The observed electroweak sector realizes it: the W couples only to left-handed fermions, and parity is maximally violated in weak processes. The chain below therefore runs from the *observed* chiral weak force — carried in the framework as H-χ' — through molecular parity violation to biological homochirality; the framework's structural contribution is the grading and the visible-sector identity, not a derivation of parity violation.

4.2 Molecular parity violation

The weak neutral current (Z boson exchange between electrons and nuclei) produces a parity-violating potential in atoms and molecules. For chiral molecules — molecules that are not superimposable on their mirror image — this creates an energy difference between left-handed (L) and right-handed (D) enantiomers [11, 12]:

$$\Delta E_{\mathrm{PV}} \sim G_F \alpha Z^5 m_e c^2 \left(\frac{a_0}{R}\right)^3$$

where G_F is the Fermi constant, Z the nuclear charge, and R the bond length. For amino acids: $\Delta E_{\mathrm{PV}} \sim 10^{-19}$ eV $\sim 10^{-17}\,kT$ at 300 K [11]. The sign is universal — it favors L-amino acids — and is fixed by the observed left-handedness of the weak coupling: given $\mathrm{SU}(2)_L$, L-amino acids are lower in energy. In the framework that left-handedness is carried as the open hypothesis H-χ' [2, §4.8], so the sign is inherited from the observed electroweak sector rather than derived from the partition.

4.3 The amplification argument

The energy difference $10^{-17}\,kT$ is far too small to directly select one handedness. The enantiomeric excess at thermal equilibrium would be $ee \sim 10^{-17}$. The question is whether amplification mechanisms can promote this bias to complete homochirality.

Three mechanisms are experimentally demonstrated. Autocatalytic amplification (the Soai reaction [13]) amplifies $ee \sim 10^{-5}$ to $> 99\%$. Crystallization (Viedma ripening [14]) drives racemic crystal mixtures to homochirality from small initial biases. Chiral polymerization with cross-inhibition [15] amplifies small monomer ee to polymer homochirality.

The gap between 10^{-17} (PVED) and 10^{-5} (Soai threshold) is 12 orders of magnitude, and bridging it from PVED alone is non-trivial. In a small prebiotic pool of N chiral molecules, random statistical fluctuations produce $ee \sim 1/\sqrt{N}$. For $N \sim 10^6$ (a microdroplet), statistical $ee \sim 10^{-3}$ — within the Viedma threshold. The PVED's structural role is to fix the *sign* of the energy difference between L and D enantiomers; how this sign propagates to biological homochirality

through the amplification mechanisms cited above is an active research question in origins-of-life. With current PVED magnitude ($\sim 10^{-17}$ ee) versus statistical fluctuations in prebiotic pools ($\sim 10^{-3}$ ee for $N \sim 10^6$ monomers), a single-pool calculation gives an L-preference probability of only $\sim 10^{-14}$ above 50%. Delivering global biological homochirality from this requires either (i) cross-pool selection mechanisms that propagate small per-pool L-biases globally, or (ii) extended timescales of prebiotic chemistry with cumulative bias amplification, or (iii) other mechanisms not yet fully characterized in the literature [13, 14, 15]. The framework's input to this chain is the PVED sign — structural only conditional on H-χ' [2, §4.8], otherwise inherited from the observed weak sector; the bridge to biological homochirality requires additional empirical mechanisms that remain partially open.

4.4 The structural conclusion

If the amplification chain is robust — every step being known physics or experimentally demonstrated, with the first link carried as the open hypothesis H-χ' — then the *sign* of the molecular parity-violating energy difference is fixed by the observed chirality of the weak coupling (conditional identification: Theorem 13 of [SM §4.8]); biological homochirality plausibly inherits this sign through amplification mechanisms whose details remain partially open in origins-of-life research (§4.3). The handedness of every amino acid in every protein in every organism is *consistent* with this sign result; whether it is *uniquely traceable* to (S, φ) depends on H-χ' and on which amplification mechanism actually operated in Earth's prebiotic chemistry.

Step	Content	Status
1	Chiral $SU(2)_L$	Observed; carried as hypothesis H-χ' [2, §4.8]
2	$SU(2)_L \to$ electroweak PV	Standard consequence
3	Electroweak PV $\to$ PVED	Known physics [11, 12]
4	Statistical fluctuation + directional bias $\to$ homochirality	Demonstrated mechanisms [13, 14, 15]; sign from Step 1
5	Biological homochirality	Observed

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5. Implications

5.1 The dissolution of fine-tuning

The fine-tuning problem assumes two freedoms: the laws could have been different, and the parameters could have been different. On the selected branch the OI framework removes the first: the structure is fixed by the stated hypotheses rather than varied. §3 shows the second is not as constrained as traditionally

claimed: the viable parameter fraction is $\mathcal{O}(10\%)$, not $\mathcal{O}(10^{-50})$. The apparent fine-tuning was dominated by structural variation that the selected branch excludes; embedded observation alone does not exclude it (§3.4).

The representative freedoms in [33, §4, Theorem 24] preserve specified data at a fixed visible/hidden interface, not every observable of every physical implementation: uniform hidden relabelling preserves the coherent channel under its stated preparation, while fibre-dependent relabelling preserves rooted populations but need not preserve coherences; an independently prepared, exactly decoupled finite ancilla also preserves that channel. These results do not show that every chemical consequence of §2 is invariant under every such transformation or that the full set of physically admissible substrata is classified. On the selected branch, the structural chain of §2 depends on A4, the continuum conditions and its stated matter, spin, statistics and atomic premises; §3's $\mathcal{O}(10\%)$ viable fraction depends on its parameter ranges and weighting.

5.2 The dissolution of the anthropic principle

The anthropic principle assumes there is an ensemble of possible universes from which observation selects. Under the empirical and structural inputs of [33, §3], including A4's lattice premise and the stated continuum and nearest-neighbor conditions, the reconstruction fixes the local lattice/gauge residue, unique up to gauge within that class [33, Theorem 23]; it does not show that embedded observation alone uniquely determines all physical implementations or eliminates structural alternatives outside that class. The preconditions for organic chemistry — atoms, carbon bonding geometry, the scale hierarchy, water, the thermal window and chiral selection — retain the additional conditions of §2. The $\mathcal{O}(10\%)$ viable parameter fraction is conditional on the ranges and weighting of §3.

A closely related concern — the Boltzmann-brain problem, where thermal fluctuations in a long-lived finite universe should overwhelmingly produce observers with random memories — is also dissolved at the framework level, with C2 read in its timescale form, by a theorem rather than by an anthropic filter. The structural observer-selection theorem [1, §4.6] shows, under its effective-mixing hypothesis, for sufficiently fast-mixing equilibria ($\tau_{\mathrm{mix}} \ll \tau_S$) and for the coarse-grained description generated by iterating the one-step effective channel, that C1–C4-satisfying observers, with C2 read in its timescale form, cannot exist in the equilibrium phase of (S, φ): the hidden-sector correlation time τ_B is bounded by the local mixing time on equilibrium microstates, so C2 fails there in its timescale form, record persistence being degraded rather than shown to fail, and — by the fast-bath erasure lemma [Main §2.3] — the readback gap of C4, the condition the characterization requires, is suppressed exponentially in τ_S/τ_B at every order $k \geq 2$. Boltzmann brains are by construction local fluctuations in a thermally equilibrated environment, so their τ_B is bounded by local thermal decorrelation and is at least six orders of magnitude too short for C2's timescale form when the bath's correlation time is set by the thermal scale —

at most $\sim 10^{-12}$ s at temperatures warm enough to support structured matter, against $\tau_S \gtrsim 10^{-6}$ s for any currently envisioned brain-like observer [Main §4.6]; the estimate fails for protected slow modes. In that description their readback is at most exponentially suppressed at orders $k \geq 2$, and they are not observers in the sense of C1–C4 with C2 read in its timescale form. The "past hypothesis" of low initial entropy — usually imported as a separate axiom to explain why we observe a low-entropy universe — is replaced, in its role of selecting observer-compatible configurations, by the same theorem, since, in that description and with C2 read in its timescale form, observers exist only in the non-equilibrium phase.

The transformations exhibited in [33, §4, Theorem 24] preserve only the observation data stated for each fixed interface and preparation. Whether additional measurements or different interfaces distinguish further implementations, and whether the physically admissible equivalences are exhaustively classified, remain open. The chemical preconditions in §2 are conditional consequences of the selected structural and matter premises, not an all-observations gauge-classification theorem.

5.3 The autocatalytic argument

On the selected branch, the structural chain supplies the preconditions for life. This section argues that autocatalytic sets are statistically expected within Kauffman's random-catalysis model, given that structure and the model's inputs.

Combinatorial diversity is structural. Carbon's sp^3 hybridization in d = 3 (§2.3) produces chain-forming chemistry with four bonding directions. The number of distinct molecules constructible from C, H, O, N grows exponentially with chain length: $\sim 20^N$ for amino acid sequences, $\sim 4^N$ for nucleotide sequences. For chains of length 50: $\sim 10^{65}$ possible peptides, $\sim 10^{30}$ possible RNA sequences. This combinatorial explosion is structural — it follows from the orbital algebra in d = 3 with four valence electrons. No other element in the derived periodic table produces comparable diversity: silicon has similar valence but weaker bonds (~ 2.3 eV for Si-Si vs. ~ 3.6 eV for C-C), insufficient for stable long chains at thermal window temperatures.

The autocatalytic threshold. Kauffman's theorem [16]: in a network of N molecule types where each molecule has probability p of catalyzing any given reaction, autocatalytic sets — closed networks that collectively catalyze their own production from simple feedstocks — emerge with probability approaching 1 when $N \times p$ exceeds a critical threshold of $\mathcal{O}(1)$. The framework's chemistry permits $N \sim 10^{65}$ possible 50-mers; the model's N counts co-present species. The question reduces to whether $p > 10^{-65}$ — whether catalysis is not astronomically rare among organic molecules.

Catalysis is not rare. Empirically, short random peptides show catalytic activity at rates suggesting $p \sim 10^{-9}$ to 10^{-7} per pair [17]. RNA fragments

catalyze their own splicing and polymerization [18, 19]. Simple amino acids catalyze aldol reactions (the Hajos-Parrish reaction). Even individual metal ions in aqueous solution show catalytic activity for a range of organic reactions. With $p \sim 10^{-9}$ and $N \sim 10^{65}$: $N \times p \sim 10^{56}$, a nominal margin of 56 orders of magnitude that counts every possible 50-mer as present.

The role of chirality. The PVED (§4) selects L-amino acids, halving the combinatorial space and — critically — eliminating cross-chiral parasitic reactions. In a racemic mixture, L-monomers can be incorporated into D-chains and vice versa, poisoning both. Homochirality, driven by the PVED's directional bias (§4.3), ensures that the combinatorial exploration proceeds within a self-consistent chemical subspace. This is not a minor refinement — it may be the difference between a productive autocatalytic network and a parasitized one [15].

What's structural vs. parametric.

Element	Status
Combinatorial diversity ($N \sim 20^L$)	Structural (carbon in d = 3)
Energy-driven exploration	Conditional (thermal window, given the inputs of §2.7)
Chirality eliminates cross-inhibition	Conditional on H-χ' (PVED sign, §4)
Catalytic probability p	Parametric (depends on bond energies, activation barriers)
$p > 1/N$	Nominally satisfied by ~56 orders (all possible 50-mers counted)

The framework derives the nominal N (structural), the conditions for a thermal window (whose existence depends on the coupling values, masses, activation energies and solvent of §2.7), and the chiral bias (conditional on H-χ', §4). The catalytic probability p is parametric but empirically enormous relative to the threshold. Within Kauffman's model, autocatalytic sets are therefore statistically expected wherever enough of that diversity is co-present; template self-replication needs the further premises of §5.4 and §5.6.

5.4 Self-replication as read-write cycling

The structural parallel. The framework attributes the memory-bearing observable statistics to read-write cycling: the observer reads the visible sector and writes correlations into the hidden sector through the coupling H_{VB}; the hidden sector stores those writes and returns them on subsequent reads; the resulting statistics are the ones quantum mechanics represents [33, §2]. Template replication is the same structural pattern at the molecular level: a polymer reads its own sequence (through complementary hydrogen bond pairing) and writes a

copy (through catalytic polymerization). Self-replication is not a special biological capability — it is molecular read-write cycling, the chemical instantiation of the same information-theoretic pattern that produces the memory-bearing dynamics quantum mechanics represents.

Von Neumann's threshold. Von Neumann's self-reproducing automaton theorem [20] establishes that in any computational system above a complexity threshold, self-reproducing configurations must exist. The threshold requires three capabilities: (a) a universal constructor — a subsystem that reads instructions and assembles a product; (b) a copying mechanism — that duplicates the instructions; (c) a control mechanism — that sequences construction and copying.

The derived chemistry provides (a) and (b); (c) is an environmental condition:

(a) Universal construction. Any catalytic polymer that reads a template sequence and assembles a corresponding product from a monomer alphabet. The autocatalytic argument (§5.3) establishes that catalysis is statistically abundant ($N \times p \sim 10^{56}$). Template-directed assembly follows from complementary pairing (structural, §2.6) combined with catalytic activity.

(b) Instruction copying. Template replication — a polymer directing the synthesis of its complement through complementary pairing. Three independently structural capabilities whose intersection is template replication (§5.5, below).

(c) Control. Environmental cycling — temperature fluctuations, wet-dry cycles, concentration gradients — drives alternation between construction and copying phases. It is a further environmental condition: the thermal window (§2.7) allows chemistry at the prevailing temperature but does not by itself supply fluctuations on timescales relevant to chemistry.

The combinatorial margin of §5.3 is Kauffman's Np, not a measure of von Neumann's requirements: it bears on the abundance of catalysis in (a), not on whether a constructor, a copier and a controller are realized together. Given environmental cycling of the kind in (c) and a chemistry that realizes (a) and (b), self-reproducing molecular configurations are possible; that they arise is a chemical premise of the argument, not a consequence of the margin.

5.5 From template replication to evolution

Template replication requires three capabilities, each independently structural: (i) *Linear information-carrying polymers* — carbon's sp^3 hybridization allows chain formation carrying N units of sequence information (structural). (ii) *Complementary pairing* — hydrogen bonding in $d = 3$ is directional, enabling selective donor-acceptor pairing between complementary sequences — the structural basis of Watson-Crick pairing (structural). (iii) *Catalytic copying* — template-directed catalysis is a subset of the statistically abundant catalysis established in §5.3, with selectivity provided by complementary pairing.

The RNA precedent. RNA is simultaneously a linear polymer, a substrate for complementary pairing, and a catalyst (ribozymes [18]). RNA-catalyzed RNA polymerization has been demonstrated [19]. This dual capability is the natural intersection of the three structural capabilities in a single molecular family.

Heritable variation is automatic. Template copying with finite fidelity produces errors. Errors in a self-replicating polymer are mutations. Faster replicators outcompete slower ones (selection), copying errors produce variants (variation), and successful variants propagate (heredity). This is Darwinian evolution — the inevitable dynamics of imperfect template replication.

The structural status of evolution.

Element	Status
Linear polymers	Structural (carbon sp^3 in $d = 3$)
Complementary pairing	Structural (H-bonding geometry in $d = 3$)
Catalytic activity	Statistically expected (§5.3)
Template replication	Intersection of the above three
Copying errors	Inevitable (finite-fidelity copying)
Selection	Inevitable (competition for finite resources)
Darwinian evolution	Consequence of replication + variation + selection

Once template replication is established, each later row follows from the preceding rows. Whether the three capabilities combine into a replicator is a chemical premise (§5.4, §5.6).

Caveats. The argument establishes that Darwinian evolution follows wherever template replication is established in a system with carbon chemistry, a thermal window and a chiral aqueous environment. It does not determine: the specific molecular system that first replicates (RNA, PNA, or something else), the timescale for the transition (millions or billions of years), or the specific pathway from simple replicators to cellular organization. These depend on the specific φ, local geochemistry, and contingent history.

5.6 The origin of life: C1–C4 at the molecular scale

The OI framework's characterization theorem is scale-independent: whenever a fast subsystem's hidden-sector coupling realizes non-trivial coupling, record persistence, sufficient capacity, and history readback (C1–C4), the subsystem's dynamics is — in the memory-bearing sector those conditions characterize — non-Markovian — history-dependent. At the cosmological scale, this produces the memory-bearing dynamics quantum mechanics represents. At the molecular scale, it produces *heredity*: a chemical system whose future depends on its past

through information stored in a persistent template. In a template replicator the four conditions take a hereditary form; satisfying them does not mark the transition to life, since non-replicating molecules such as single proteins also satisfy them (§7.2).

Before life: Markovian chemistry. Ordinary chemical reactions are Markovian — the reaction rate depends on current concentrations alone, not on the history of the mixture. Products do not remember how they were made. Each reaction cycle is statistically independent of the previous one. No memory, no heredity, no evolution.

The transition: molecular C1–C4 in a replicator. In a template replicator the four conditions take chemistry-level forms: (C1) coupling, realized as catalytic feedback — the template directs synthesis and synthesis maintains the template; (C2) persistence — the template outlives individual reaction cycles (RNA half-life of hours–days vs. reaction times of seconds–minutes); (C3) capacity — the template's sequence space is large enough to encode catalytic function ($4^{20} \sim 10^{12}$ for a 20-mer RNA, vastly exceeding the $\sim$ 10–100 reactions in a minimal metabolism); (C4) history readback — copying reads the stored sequence back, so the system's next state depends on its history beyond its present coarse composition. Catalytic feedback, template copying and growth are properties of the replicator, not consequences of C1–C4.

After life: non-Markovian chemistry. The system's future depends on its history, stored in the template. This is heredity — the defining property of life — derived from the same C1–C4 conditions that diagnose an accessible non-Markovian realization at the cosmological horizon.

RNA as a single-molecule template and catalyst. RNA is the natural polymer known to serve as both template (persistent; 4-letter alphabet with arbitrary length) and catalyst (ribozymes catalyze RNA polymerization [18, 19]) in a single molecular species. DNA stores information but cannot catalyze; proteins catalyze but cannot easily template. The distinction concerns that dual role, not C1–C4: single proteins satisfy C1–C4 as memory conditions (§7.2) without templating. The RNA world hypothesis rests on this chemical evidence; it is an empirical hypothesis, not a consequence of C1–C4, and §5.5 leaves open whether RNA, PNA or another polymer replicated first. DNA and proteins are later optimizations: DNA improves C2 (greater chemical stability), proteins improve C1 (more diverse catalysis). The RNA $\rightarrow$ DNA+protein transition refines C1–C4 without changing the qualitative architecture.

Replicators under population dynamics. C1–C4 do not by themselves imply copying or growth: the finite bijections that realize them in [1, §3.4] preserve the number of states and neither copy nor grow. Given a template replicator in a feedstock-rich environment, with replication exceeding decay and copying fidelity high enough to keep variants heritable, standard population dynamics gives exponential growth while feedstock lasts, and the replicator can come to dominate its environment. Exponential growth is not peculiar to replicators:

mass-action autocatalysis also grows exponentially while its feedstock lasts. The persistence of a replicator population is conditional on sustained feedstock and on replication exceeding decay. Whether a replicator arises in a given aqueous organic chemistry depends on chemical premises — catalytic closure that includes template copying, co-presence of the relevant species, environmental cycling — that the autocatalytic argument (§5.3) addresses only within Kauffman's model and its inputs.

The timescale. The earliest evidence for life on Earth is ~ 3.8 Gya (carbon isotope signatures in Isua, Greenland [32]), on a planet that formed ~ 4.5 Gya. The ~ 700 Myr gap is consistent with a period of prebiotic chemical exploration before the first replicator, whose duration depends on solution-specific parameters (concentration, temperature fluctuation rates, mineral surface availability).

5.7 From evolution onward: companion to *Computation.md*

The structural chain established through §5.6 reaches biological evolution as a natural terminus. The next link — from evolved biological observers to information-processing systems, neural computation, semiconductor-based universal computation, and the embedded-computation theory that follows — is the subject of the companion paper *Embedded Computation in the Incompleteness of Observation Framework* [38]. There, the chain is continued through general intelligence, AI as embedded-observer construction, the Gödel-Turing-OI incompleteness family applied recursively, the BQP characterization theorem for embedded observers, and the framework's position on the P vs NP question. The dividing line between this paper and the companion reflects the qualitative shift from *material-structural* consequence-tracing (chemistry-biology) to *informational-structural* consequence-tracing (computation-complexity).

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6. The Existence Question

6.1 The last question

The structural chain derives everything from (S, φ) — physics, chemistry, life, intelligence, AI. But it does not derive (S, φ) itself. The reconstruction theorem [33, §3] establishes, under its stated inputs within the nearest-neighbor class:

$$\text{Observed physics} \quad \longleftrightarrow \quad [(S, \varphi)]/\sim$$

This is conditional: *if* observation exists, *then* (S, φ) exists. The question is whether the framework can say anything unconditional about *why* (S, φ) exists at all.

6.2 Mathematical existence

(S, φ) is a finite set and a bijection. This is a mathematical object — a structure in the space of logical possibilities. It requires no physical substrate, no creator, and no cause. It exists in the same sense that the integers exist or that the permutation group S_n exists: as a logically consistent structure that cannot fail to be well-defined.

If mathematical structures are taken as necessarily existent — the Platonic position — then (S, φ) *necessarily* exists. Not "happens to exist" or "was created" but cannot fail to exist, because there is no logical contradiction in its definition. On this reading, "why is there something rather than nothing?" dissolves: the question assumes non-existence is the default and existence requires explanation. For mathematical structures, existence is the default. Non-existence would require a logical inconsistency, and (S, φ) has none.

6.3 Observable mathematics

The framework makes this sharper than generic Platonism. Tegmark's Mathematical Universe Hypothesis [7] asserts that all consistent mathematical structures are equally real. The OI framework says something more specific: among all mathematical structures, only those containing embedded observers are *observable from within*. The characterization theorem [1, §3.4] identifies which structures contain embedded observers: exactly those of the form (S, φ) with conditions C1–C4. The reconstruction theorem [33, §3] fixes the local lattice/gauge residue of the framework's physics under its stated inputs; it does not select a unique Bell-inclusive completion.

At the level Theorem 23 actually controls, the structural inputs, within the nearest-neighbor class, select one local lattice/gauge residue. This does not imply one unique Bell-inclusive substratum: preparation-indexed Bell-composite structure remains underdetermined even when H-Bell compatibility holds. The framework's mathematical-existence discussion therefore concerns the existence of admissible deterministic structures, not uniqueness of the full ontology.

6.4 The self-referential closure

The complete chain:

$$(S, \varphi) \text{ exists (mathematical structure)}$$
$$\xrightarrow{\text{conditional}} \text{contains embedded observers seeing QM, GR, SM}$$
$$\xrightarrow{\text{conditional; coupling values, masses}} \text{orbitals, periodic table, carbon, scale hierarchy, water, thermal-window conditi}$$
$$\xrightarrow{\text{statistical}} \text{autocatalytic networks (nominal } N \times p \sim 10^{56})$$
$$\xrightarrow{\text{conditional; constructor, copier, cycling}} \text{self-reproducing configurations}$$

$$\xrightarrow{\text{structural}} \text{template replication} \xrightarrow{\text{inevitable}} \text{Darwinian evolution}$$

$$\xrightarrow{\text{generic}} \text{information processing} \xrightarrow{\text{contingent}} \text{general intelligence}$$

$$\xrightarrow{\text{material with suitable gap}} \text{semiconductors} \xrightarrow{\text{theorem}} \text{universal computation} \xrightarrow{\text{structural}} \text{AI}$$

$$\rightarrow \text{observers ask: “why does } (S, \varphi) \text{ exist?”}$$

The chain combines conditional physical results, statistical models and contingent biological or computational steps. Its chemical starting point inherits the selected lattice/continuum branch and the stated matter assumptions; later conclusions retain their own premises. General intelligence is one of the contingent steps.

6.5 What cannot be answered

One question is outside the framework's scope — not contingently but necessarily:

Why is there mathematical structure at all? The framework takes mathematical existence as given. It does not explain why logical consistency is a property that structures can have, or why “exists” and “is logically consistent” are related. This is not a gap in the framework — it is the boundary of all possible frameworks. Any explanation of mathematical existence would itself be a mathematical structure, requiring explanation in turn. The regress has no bottom.

The existence question is distinct from the conditional physical and biological programme developed here. The named lattice, continuum, matter and operational bridge assumptions remain open hypotheses within that programme, and the biological extensions retain their own model assumptions.

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7. Quantum Biology: C1–C4 Inside Living Systems

7.1 The reframing

Quantum biology — the study of quantum coherence in biological systems — is an active and contested field. The central question is usually framed as: “How do biological systems maintain quantum coherence at warm, wet, noisy conditions?” The framework reframes this. QM is not a fragile state that biology must protect from decoherence. The memory-bearing non-Markovian dynamics — universally quantum-representable — is the necessary form for any subsystem whose hidden-sector coupling realizes C1–C4 (the characterization theorem [1, §3.4]). If biological subsystems realize C1–C4 internally, their dynamics is memory-bearing — accessibly non-Markovian, with P-indivisibility the stronger diagnostic requiring a measured witness, not automatic (`review4_probes.py`)

— not because evolution engineered quantum coherence, but because the coupling architecture mandates it.

7.2 C1–C4 in proteins

Consider an enzyme's active site as the visible sector and the protein scaffold's conformational degrees of freedom as the hidden sector.

C1 (coupling). The active site is covalently bonded to the scaffold — coupling is continuous and bidirectional. Electronic transitions at the active site induce conformational strain in the scaffold; conformational changes in the scaffold modulate the active site's electronic structure. This is the standard picture of allostery.

C2 (memory persistence). Electronic transitions at the active site occur on femtosecond timescales ($\tau_S \sim 10^{-15}$ s). Conformational changes of the protein scaffold occur on microsecond to millisecond timescales ($\tau_B \sim 10^{-6}$ to 10^{-3} s). The ratio: $\tau_S/\tau_B \sim 10^{-12}$ to 10^{-9}. This is the *inverse* of the Markovian regime — exactly the slow-bath realization of persistence. The scaffold retains correlations for $\sim 10^9$ electronic transitions before relaxing.

C3 (capacity). A typical enzyme has $\sim 10^3$ to 10^4 atoms, each with multiple conformational degrees of freedom. The conformational state space is exponentially large: $\sim q^{N_{\text{conf}}}$ with $N_{\text{conf}} \sim 10^3$. This exceeds the active site's electronic state space (~ 10 to 100 relevant states) by tens of orders of magnitude.

C4 (history readback). The stored conformational strain modulates subsequent active-site transitions — the coupling reads back what it wrote; readback, not storage alone, is the load-bearing memory condition (`review4_probes.py`).

All four conditions are realized. What the characterization theorem licenses: the active site's reduced dynamics is accessibly non-Markovian — memory-bearing — with P-indivisibility the stronger diagnostic requiring a measured witness.

A scope note is essential here. The characterization theorem is proved for a *closed, reversible* substratum (a bijection on a finite configuration space; [Main] §2.3). An enzyme and its scaffold are neither closed nor reversible — the scaffold thermalizes with solvent, and phase-space volume contracts as heat leaves — so the enzyme-over-scaffold trace-out is not itself an instance of the theorem. What transfers rigorously is the model-independent consequence (below): coupling that writes to, and reads back from, a persistent high-capacity sector yields non-Markovian, history-dependent dynamics. Genuine P-indivisibility is a property of the *closed substratum* of which the enzyme is a partition; its observable magnitude at the molecular scale is not established here. The predictions of §7.3–§7.4 should therefore be read as tests of molecular *memory*, consistent with the framework, rather than as isolating P-indivisibility from classical conformational memory.

7.3 The prediction

The protein scaffold is a slow bath. Correlations written into the scaffold during one catalytic event persist through subsequent events. The active site's transition probabilities are history-dependent. This produces:

History readback. The scaffold returns correlations to the active site on conformational timescales (μs to ms), modulating catalytic rates in a way that depends on the enzyme's recent history. Standard rate theory predicts exponential kinetics; this memory predicts non-exponential kinetics with a specific signature: the rate "constant" oscillates or shows memory effects on the conformational timescale.

Non-Markovian signatures. The mutual information $I(X_{<t}; X_{>t}|X_t)$ between past and future catalytic events, conditioned on the present state, is positive wherever the scaffold's record is read back: by the readback lemma [1, §2.3], a readback gap δ at a history of probability p_0 gives $I(X_{<k}; X_{k+1} \mid X_k) \geq p_0\delta^2/\ln 2$, and the past–future quantity is at least as large. Its lower bound is set by δ and p_0, which are open for the protein system.

7.4 Existing evidence

Single-molecule enzyme kinetics. Individual enzyme molecules show fluctuating catalytic rates with correlations persisting for milliseconds to seconds (English et al. 2006 [21]; Lu et al. 1998 [22]) — precisely the signature of a slow bath modulating active site dynamics. The multi-timescale structure has been characterized at single-molecule resolution for T4 Lysozyme [34], where three distinct conformational states with ns-to-ms exchange times have been resolved, each playing a different role in the catalytic cycle.

Dispersed kinetics. Many enzymes show stretched-exponential waiting-time distributions rather than the single-exponential predicted by Markovian theory — generic in memory-bearing systems. The trace-out structure has been formalized within the single-molecule biophysics literature [35]: low-dimensional projections of high-dimensional molecular dynamics generically exhibit memory effects and produce non-Markov processes, independent of the specific dynamical model.

Photosynthetic energy transfer. Long-lived coherences in the FMO complex were first reported by Engel et al. [23]. Subsequent work (Duan et al. 2017 [36]; Cao-Plenio 2025 [37]) has refined the interpretation: the long-lived coherences are vibronic — superpositions of electronic exciton states with the protein's vibrational environment — rather than purely electronic. Purely electronic coherences dephase within ~240 fs; the picosecond-scale coherences observed in 2D spectra arise from coupling between the electronic states and the slow vibrational protein scaffold. This is the C1–C4 structure operating: the protein scaffold (slow bath, C2) mixes with the electronic transitions at FMO chromophores (fast, visible) to produce coherences that persist for $\sim \tau_B$ rather

than τ_S. The picosecond coherence timescales correspond to vibrational mode periods of the protein scaffold, consistent with the framework's prediction that protein-scale C1–C4 systems exhibit dynamics on the slow-bath timescale.

Cryptochrome radical-pair magnetoreception. Avian magnetic compass orientation operates through cryptochrome radical pairs in the retinal photoreceptors. Radiofrequency fields at the electron Larmor frequency disrupt magnetic orientation in migratory birds (Wiltschko-Wiltschko 2019 [39]), diagnostically confirming radical pair mechanism. The crucial radical pairs have millisecond lifetimes [40]. Millisecond coherence in a protein-embedded quantum system at warm physiological conditions sits at the upper end of the protein conformational τ_B range identified in §7.2. The avian compass is therefore an operating biological quantum sensor with coherence times the framework predicts as structurally required by C1–C4 inside protein scaffolds.

Conformationally gated tunneling. Hydrogen tunneling in enzymes shows temperature-independent kinetic isotope effects [24] — inconsistent with transition state theory but consistent with quantum tunneling modulated by slow conformational dynamics (C2).

7.5 Implications for drug design

Allosteric drugs. The framework predicts that allosteric effects carry temporal correlations through the scaffold — the drug's binding event writes conformational information that is returned to the active site over the conformational timescale. Optimal design should account for these temporal correlations, not just equilibrium binding affinities.

Drug resistance. Resistance mutations often occur far from the active site. The framework predicts they cluster in regions that maximally alter the scaffold's slowest conformational modes — altering the *memory structure* of the hidden sector. Testable via normal mode analysis.

Enzyme engineering. The framework suggests optimizing the *coupling architecture* (C1–C4 between active site and scaffold), not just the active site's electronic structure. Better-tuned τ_S/τ_B ratios should yield both higher catalytic rates and more robust performance.

7.6 Epistemic status

The application of C1–C4 to protein systems rests on the model-independent consequence of §7.2: write-and-readback coupling to a persistent high-capacity sector yields history-dependent dynamics. The enzyme-over-scaffold trace-out is open and dissipative, so the characterization theorem itself does not apply to it (§7.2, scope note). The specific signatures (non-exponential kinetics, fluctuating rates, conformational gating of tunneling) are consistent with existing experimental data but have not been quantitatively tested against the framework's specific predictions (positive $I(X_{<t}; X_{>t}|X_t)$ at accessible orders, history-dependent

catalytic rates). The medical implications (allosteric temporal correlations, resistance as memory-structure alteration) are consequences of the prediction but have not been independently tested.

The summary: the framework expects enzyme kinetics to be non-Markovian wherever the scaffold writes and reads back catalytic history. The prediction is consistent with existing evidence and makes specific testable claims. Whether the non-Markovian corrections are large enough to matter for practical drug design is an empirical question. The ratio $\tau_S/\tau_B \sim 10^{-12}$ to 10^{-9} fixes how long the scaffold retains a record, of order 10^9 to 10^{12} active-site transitions (§7.2), not the size of the correction, which is set by how strongly the record modulates the active site (the gap δ of §7.3) and is open for the protein system.

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8. The GPS Analogy: Where OI Corrections Reach Engineering Relevance

8.1 The precedent

Nobody building a satellite navigation system in the 1950s would have predicted they would need general relativity. GPS requires relativistic corrections because satellite clocks experience different gravitational potentials — 45 μs/day from GR, 7 μs/day from SR. Without corrections, position drifts ~10 km/day. The practical application emerged when engineering precision reached the scale where the correction mattered.

The framework's specific correction is non-Markovian dynamics: deviations from standard Markovian models wherever a bath retains a record and reads it back. Their size is set by the coupling and the readback, not by τ_S/τ_B; the ratio sets how long the record is retained, of order τ_B/τ_S system events under uniform erosion [1, §2.3]. At the cosmological scale, $\tau_S/\tau_B \sim 10^{-32}$. But τ_S/τ_B is scale-dependent. The question: where does engineering precision reach the scale where OI corrections matter?

8.2 Quantum computing

The Markovian assumption in error correction. Standard quantum error correction (surface codes, stabilizer codes) assumes Markovian noise — each error is statistically independent of previous errors. The framework predicts this is wrong whenever the qubit's environment satisfies C2 (memory persistence).

The physical system. Superconducting qubits (transmons) are coupled to two-level systems (TLS) in the substrate — structural defects in the amorphous oxide layer. TLS dynamics is slow ($\tau_B \sim \mu$s) relative to gate operations ($\tau_S \sim$ ns). The ratio: $\tau_S/\tau_B \sim 10^{-3}$. This is 29 orders of magnitude larger than the cosmological case.

The prediction. The noise is memory-bearing: error probabilities at gate k depend on the error history at gates $k-1, k-2, ...$ through correlations stored in the TLS bath. Standard Markovian error correction sees these correlated errors as an anomalously high uncorrectable error rate. Non-Markovian error correction protocols — designed for the actual correlated noise structure — could exploit the correlations rather than treating them as random, improving logical error rates.

The scale. The ratio fixes the correlation length, of order $\tau_B/\tau_S \sim 10^3$ gates, not the per-gate size of the correction, which depends on the qubit–TLS coupling and is not derived here. Circuits of 10^3 gates or more span the bath's correlation time; how much the correlated component adds to the error budget over 10^4 to 10^6 gates depends on that size and on whether the correlated errors add coherently.

A decoder-level test on existing data. P-indivisibility has a syndrome-level signature. Three grades of temporal structure are distinguished: (i) non-stationarity (parameter drift — classical and divisible), (ii) correlated noise that is non-Markovian while its population law stays divisible (leakage, cosmic-ray events), and (iii) P-indivisible noise, witnessed by information backflow in the population law or by a negative divisor — non-monotonic conditional error-rate relaxation and negative canonical decay rates in the per-round channel reconstructed from syndrome history. Readback (C4) at an accessible order places the noise in grade (ii) or (iii); it does not select grade (iii), since a readback gap is compatible with a population law that is P-divisible on every window short of recurrence [1, §2.3]. Grade (iii) is a further property, testable on the same records by a divisibility witness. The prediction is directional and falsifiable at device level: after conditioning on drift and known classically-correlated artifacts, a history-conditioned error predictor should retain an exploitable residual kernel, and decoders conditioned on syndrome history should outperform memoryless decoders by a margin set by that kernel. This is testable on already-public surface-code memory-experiment datasets (distance-3/5/7 with full detection-event records). The claim at device scale concerns the universality of C3 memory in the engineered bath (two-level systems and slow modes); the bath-motion part of the cosmological residue $\mathcal{D}$, of order 10^{-64} [1, §3.2], lies some sixty orders below device noise and is not at issue in this test.

Status of the syndrome-level test (carried out). This test has been run against Google's Willow surface-code detection-event streams across distances 3, 5, and 7 in both bases, and its corrected claims ladder is recorded here in full, including the rungs that were withdrawn or revised. An initial pass reported per-experiment positives, but its null lacked a measurement-error control and detected only known-Markovian structure; those verdicts are *withdrawn*. A second pass reported nineteen of 336 experiments positive, but under a stringent empirical null built from each experiment's own detector error model, none survive multiple-comparison control — the nominal per-experiment significances were overdispersed at the real geometries and are not calibrated, so *no individual-*

experiment detection is defensible. What does survive is a population-level trend: a temporal memory beyond the detector error model whose lag-≥ 2 exploitability rises systematically with cycle depth (mean Spearman $\rho = +0.74$ across 28 series — imperfectly independent given the shared device — with per-series permutation $p \approx 5 \times 10^{-5}$) and, more weakly across the three code distances, with distance; auxiliary long-memory (non-saturating windowed-rate variance) and shot-scale non-stationarity corroborate it. This establishes that beyond-model history dependence exists and accumulates with depth — the *syndrome-testable* half of the prediction — but not its attribution. No information-backflow signature is present anywhere; however, the projective measurement cycle is, by the framework's own τ_S/τ_B relation, structurally insensitive to coherent revival at the dominant bath timescale, so this null neither supports nor excludes the P-indivisibility mechanism — the syndrome venue cannot decide it, and the timescale constraint, which was in hand before the test, should have located the informative venue (free evolution, §8.3) in advance. The attribution of the surviving memory between benign carriers (heating and leakage accumulation, $1/f$ rate modulation — divisible, grade-ii) and any genuine P-indivisible component is undecidable from syndrome streams and remains open.

The net device-level status is therefore: the framework's syndrome-testable prediction — that error probabilities carry exploitable history dependence beyond the detector model, growing precisely in the large-code, long-run regime fault-tolerant hardware is now entering — is supported and is engineering-relevant; the framework-distinctive prediction, that this memory is specifically P-indivisible, is not decided by this venue and its backflow proxy returned null. The claim is consistent with the data, unfalsified, and at the phenomenology level evidentially inert: the observed memory is degenerate with standard non-Markovian open-systems physics. The framework's distinctive exposure lies not in this phenomenology but in cross-platform kernel-class universality — and in the free-evolution sensing venue of §8.3, not the projective syndrome venue tested here.

8.3 Quantum sensing

Nitrogen-vacancy (NV) centers. NV centers in diamond detect magnetic fields with sensitivity approaching the standard quantum limit, bounded by the coherence time T_2. Standard models assume Markovian decoherence — coherence decays exponentially and is gone.

The OI prediction. The diamond lattice's phonon bath has slow modes (paramagnetic impurities, strain fields with relaxation times ~ms–s). C2 is satisfied: τ_S (electronic spin dynamics, ~ns) $\ll \tau_B$ (bath relaxation, ~ms). The decoherence is non-Markovian; where the bath returns its stored correlations, coherence partially revives at times set by the bath's correlation time. That revival is information backflow: correlations written into the lattice during one measurement are returned during a later measurement.

The application. Sensing protocols that exploit the revival — measuring at the backflow time rather than during the initial decay — could push sensitivity beyond the Markovian-assumed T_2 limit. The size of the improvement is not fixed by τ_S/τ_B: it depends on the revival amplitude, which requires an NV-specific model of the bath, readout and control (§8.9).

8.4 Precision metrology and atomic clocks

Current situation. Optical lattice clocks achieve fractional frequency uncertainty $\sim 10^{-18}$ — sensitive enough that GR corrections from a 1 cm height difference are measurable. At this precision, the atoms' interaction with the lattice light field may satisfy C1–C4: the lattice photons provide coupling (C1), the optical cavity has slow modes (C2), and the photon field has high capacity (C3).

The prediction. If the atom-cavity system satisfies C2 with sufficient separation, the atomic transition dynamics is non-Markovian. Clock instability would show correlations at the cavity correlation time — not the white frequency noise assumed by standard Allan variance analysis. Whether the correction is measurable at 10^{-18} fractional uncertainty depends on the specific cavity parameters, but the framework predicts its existence as a structural consequence.

8.5 The pattern

Domain	System / Bath	τ_S/τ_B	Current relevance
Cosmology	Observer / trans-horizon	10^{-32}	Framework's home ground
Enzymes	Active site / scaffold	10^{-9} to 10^{-12}	§7: consistent with data
Quantum sensing	NV spin / lattice defects	10^{-6}	Approaching relevance
Quantum computing	Qubit / TLS	10^{-3}	At the threshold
Atomic clocks	Atom / cavity	$\sim 10^{-6}$	Potentially measurable

The ratio τ_S/τ_B does not order the size of the corrections; it orders how long each bath retains its record, from the cosmological horizon, which retains it longest, to engineered devices, whose retention windows of $\sim 10^3$ to 10^6 events fall within experimental run lengths. The framework predicts the same structural phenomenon — memory-bearing dynamics from a persistent, read-back bath, with P-indivisibility the stronger witnessed diagnostic — at every scale. The question in each case is whether the correction has reached the precision frontier.

The GPS analogy is exact: GR corrections were irrelevant until clocks got precise enough. OI corrections are irrelevant until quantum devices get precise enough. The framework predicts that as quantum technologies continue improving, non-Markovian corrections will transition from negligible to measurable to design-relevant — following the same trajectory that relativistic corrections followed from Newtonian mechanics to GPS.

8.6 Engineered partitions: the bath as a resource

Standard quantum engineering treats the environment as the enemy — decoherence destroys quantum information, and the goal is to isolate qubits from their surroundings. The framework suggests a fundamentally different design philosophy: *engineer* the C1–C4 memory architecture and realize the target quantum process as its finite-test shadow ([Main §3.4]) — the conditions do not by themselves select the operational mechanism.

The design principle. Instead of isolating a qubit from all environmental coupling, deliberately couple it to a *slow, high-capacity hidden sector* with controllable properties. The hidden sector stores correlations written during one gate operation and returns them during a later operation. The bath is not noise — it is programmable quantum memory.

Concrete platform: qubit + spin chain. A qubit coupled to a linear chain of L ancillary spins. The chain is the engineered hidden sector. Its correlation time scales as $\tau_B \sim L^z$ (where z is the dynamical exponent — for a Heisenberg chain, $z = 1$). Its capacity scales exponentially: $\sim 2^L$ states. By tuning L (the lengths below are illustrative estimates from $\tau_B \sim L^z$, not derived critical lengths):

- Short chain ($L \lesssim 5$, $\tau_B \sim \tau_S$): how far the dynamics departs from the Markov benchmark, and whether any record survives to readback, depends on the coupling, the chain's conditional mixing and its readback; the ratio τ_B/τ_S alone establishes neither memorylessness nor record survival.
- Long chain ($L \gtrsim 20$, $\tau_B \gg \tau_S$): memory-bearing regime where the chain retains its record and reads it back (C4). Correlations written at gate k can then return at later gates, within a retention time of order τ_B/τ_S gates under uniform erosion, partially restoring coherence. P-indivisibility of the qubit's population law on such windows is a further property, certified only by a divisibility witness.
- Whether a record survives to readback depends on the chain's conditional mixing, not on a threshold in τ_B/τ_S; where a record survives and the chain reads it back (C4), the visible law carries accessible finite-horizon memory. On a fixed finite reversible representative, where recurrence returns the visible map to the identity and a readback gap forces a non-permutation rooted map, that memory also shows up as stochastic indivisibility of the visible population law across the recurrence cycle [1, §2.3]. Neither is a claim about generic open-system non-Markovianity of a quantum channel.

This is testable in existing platforms: superconducting qubits coupled to engineered spin chains [25], NV centers in diamond with controllable nuclear spin baths, or trapped ions with engineered phonon modes. The prediction: where the chain retains and reads back its record (C4), the qubit's dynamics acquires accessible conditional memory, visible in history-dependent multi-time statistics; partial revivals of T_2 coherence at multiples of τ_B are a further signature, present only where the chain's return also revives distinguishability.

8.7 Non-Markovian quantum computation

The standard assumption. Quantum circuits are sequences of unitary gates, each assumed to act independently of past gates. Error correction adds redundancy to protect against independent errors. The entire architecture assumes Markovian noise.

The OI alternative. If gates share a slow bath (a common hidden sector with τ_B longer than the circuit depth $\times \tau_S$), the gates carry temporal correlations. Gate k's effect depends on what gates $k-1, k-2, ...$ did, through correlations stored in the bath.

Built-in error correction. Dynamics with information backflow *increase* the distinguishability of quantum states over some intervals, which certifies P-indivisibility, and so reverse some decoherence without requiring additional overhead. In Markovian evolution, distinguishability only decreases (data processing inequality). Where the evolution shows backflow, the bath returns information that was temporarily inaccessible. This is a form of dynamical error correction built into the physics rather than added as a computational layer.

The open question. Whether P-indivisible gate sequences can perform computations that no Markovian circuit of equal depth can is a well-posed mathematical question. The framework provides the tools — P-indivisible stochastic processes, the readback lemma [1, §2.3] — but the computational complexity implications are unexplored. If the answer is yes, it would constitute a new model of quantum computation beyond the standard circuit model, with the bath playing the role of a shared temporal resource analogous to entanglement as a spatial resource.

8.8 Quantum materials by design

Heavy-fermion materials as candidate memory-bearing systems. Heavy-fermion compounds ($CeCoIn_5$, $YbRh_2Si_2$, UPt_3) contain localized f-electrons forming a slow bath ($\tau_B \sim$ ps to ns) coupled to itinerant conduction electrons ($\tau_S \sim$ fs). The ratio: $\tau_S/\tau_B \sim 10^{-3}$ to 10^{-6}. These materials exhibit anomalous transport — non-Fermi liquid resistivity ($\rho \sim T^\alpha$ with non-integer α), logarithmic specific heat, and memory effects in magnetoresistance. The standard explanation invokes Kondo physics and quantum criticality. The OI framework provides a structural reframing: if the conduction electrons satisfy C1–C4 with the f-electron bath, their dynamics is non-Markovian on

accessible windows — the readback gap forces positive conditional memory [1, §2.3] — and the anomalous transport is then a consequence of non-Markovian electronic dynamics rather than a deviation from normal behavior requiring a special mechanism. P-indivisibility of the electronic population law is a further property, not implied by C1–C4 on those windows.

The design principle. In natural materials, τ_S/τ_B is fixed by the chemistry. In engineered metamaterials, it can be tuned. Candidate platforms: (i) *Molecular junctions* — single molecules bridging electrodes, with floppy side chains as slow bath ($\tau_S/\tau_B \sim 10^{-9}$); prediction: history-dependent conductance. (ii) *Cold-atom lattices* — two species with independently tunable tunneling rates; prediction: non-thermal momentum distributions at large τ_B/τ_S. (iii) *Photonic crystals with quantum dots* — engineered bandgap suppresses fast decay while maintaining coupling; prediction: non-exponential fluorescence with partial revivals.

8.9 Quantitative performance estimates

The error-correction and materials cases rest on hidden-sector memory, which C4 supplies on accessible windows; the coherence and sensing cases rest additionally on a revival of distinguishability, which certifies P-indivisibility on the window and which C1–C4 do not supply ([1, §2.3]; §8.8). Each is organized by the ratio τ_B/τ_S, which the framework treats as the design parameter.

Quantum error correction: a structural correlation, not a quantified gain. Standard surface codes assume independent (Markovian) errors. Correlated errors from a slow bath extend over $\sim \tau_B/\tau_S$ gates. A Markovian decoder sees these correlations as a higher effective error rate — it *underperforms*. A correlation-aware decoder exploits the predictability: correlated errors are easier to correct than random errors. For superconducting qubits with TLS baths ($\tau_S/\tau_B \sim 10^{-3}$, correlations over $\sim 10^3$ gates), a correlation-aware decoder could reduce the effective error rate at the same physical error rate. The size of that reduction, and hence any change in the physical-to-logical qubit overhead, is not quantified here: it requires a code, a decoder and a noise spectrum, none of which the framework supplies.

Coherence extension: a structural revival, not a quantified gain. Standard coherence decays as e^{-t/T_2}. Where the engineered bath's return also revives distinguishability (information backflow), coherence partially revives at multiples of τ_B. The revival amplitude per cycle is not fixed by τ_S/τ_B: it depends on the coupling and on how much of the stored record the bath returns, and it is not quantified here. Over multiple revival cycles before the bath relaxes, coherence persists longer than a Markovian estimate gives; the extension factor is not quantified.

Quantum sensing: a structural information recovery, not a quantified gain. Standard NV magnetometry sensitivity: $\delta B \sim 1/(\gamma\sqrt{T_2 \cdot T_{\text{meas}}})$. Where the bath's return at τ_B also revives distinguishability (information backflow, a

property C1–C4 do not by themselves supply), each measurement can recover information that Markovian protocols assume is lost. The size of the sensitivity gain is not quantified: it requires an NV-specific model of the bath, readout and control.

Quantum materials: qualitative regime change. Where an engineered bath with $\tau_S/\tau_B \sim 10^{-1}$ couples strongly to the conduction electrons and reads its record back, non-Markovian corrections need not be perturbative; their strength is set by the coupling, while the ratio only confines the retention window to ~ 10 electronic events. Where the corrections are $\mathcal{O}(1)$, standard band theory gives qualitatively wrong predictions, and the material's electronic properties are determined by memory-bearing (non-Markovian) dynamics, not by Markovian approximations. This is not a percentage improvement — it is a new regime of matter. The heavy-fermion precedent (non-Fermi liquid transport, unconventional superconductivity, hidden order) suggests that qualitatively new phenomena emerge when the non-Markovian corrections become $\mathcal{O}(1)$.

Domain	τ_S/τ_B	Estimated gain	Status
Error correction	10^{-3}	correlated errors over $\sim \tau_B/\tau_S$ gates; overhead gain not quantified (no code/decoder model)	Testable now with correlation-aware decoders
Coherence extension	10^{-1} (engineered)	coherence beyond the Markovian estimate; factor not quantified	Requires engineered spin chain coupling
Quantum sensing	10^{-6}	information recovered per measurement that Markovian protocols discard; sensitivity gain not quantified (no readout/control model)	Requires backflow-optimized protocols
Quantum materials	10^{-1} (engineered)	Qualitatively new regime at strong coupling	Requires metamaterial fabrication

8.10 Existing experimental evidence

The framework's predictions about non-Markovian effects are not speculative — they are corroborated by existing experimental data across multiple domains. The literature has been documenting these effects for over two decades without

a unifying structural explanation. The OI framework provides one: wherever C1–C4 are satisfied, accessible non-Markovianity is mandatory — the readback gap forces positive conditional mutual information on the accessible window [1, §2.3].

Superconducting qubits. Agarwal et al. [26] found that purely Markovian noise models cannot reproduce experimental data from driven superconducting qubits. The non-Markovian dynamics arises from two-level system (TLS) interactions in the substrate — precisely the slow bath (C2) the framework identifies. White et al. [27] performed the first full multi-time quantum process tomography on superconducting processors and found non-Markovian noise present in all cases measured, with a significant fraction originating from genuine quantum correlations across time. Notably, Burkard and collaborators [28] found that QAOA algorithm performance *improves* as the noise correlation time increases at fixed local error probability — direct evidence that correlated noise is a resource, not merely an obstacle, exactly as §8.7 predicts. More recent work has begun to operationalize this resource: Puviani et al. (Phys. Rev. Lett. 2025) [41] used reinforcement learning to train recurrent neural networks providing quantum error correction schemes based on memory — responding to the full history of measurement outcomes rather than only the most recent one — and demonstrated that these non-Markovian feedback schemes significantly outperform standard Markovian strategies. Mandayam and collaborators [42] developed Petz-recovery-map QEC schemes adapted to non-Markovian noise, with the non-Markovian-adapted Petz map outperforming standard stabilizer codes. The transition from documenting non-Markovian noise as a problem (Agarwal, White) to engineering it as a resource (Burkard, Puviani, Mandayam) is the empirical realization of §8.7's structural prediction.

Single-molecule enzymology. Edman and Rigler [29] directly measured "memory landscapes" of single horseradish peroxidase molecules, extracting non-Markovian behavior from the catalytic cycle. The enzyme's activity fluctuates over timescales from milliseconds to seconds — the signature of a slow conformational bath (C2) modulating the active site's electronic dynamics. Kou and Xie [30] showed that slow conformational interconversion produces memory effects in successive enzymatic turnover times: the waiting time for turnover $k + 1$ is correlated with the waiting time for turnover k, with correlation strength decaying on the conformational timescale. This is precisely the conditional memory predicted by the readback lemma [1, §2.3] applied to the protein system.

The unifying explanation. These experimental results — from quantum computing hardware and from single-enzyme biophysics — are conventionally treated as unrelated phenomena requiring separate theoretical frameworks. The OI framework unifies them: both are instances of memory-bearing dynamics arising from a slow, high-capacity hidden sector coupled to a fast observable subsystem. The TLS bath in a superconducting chip and the conformational bath in a protein scaffold play the same structural role — they satisfy C2 (persistent — slow in both realizations) and C3 (sufficient capacity), and where their records

are read back (C4) they produce the same qualitative phenomenon (memory effects, non-exponential dynamics) at different scales.

8.11 Cross-domain observation: universal memory strength in single-entity systems

The non-Markovian dynamics documented in §8.10 is conventionally quantified by domain-specific measures. These can be converted to the Hurst exponent H ($H = 0.5$: Markovian; $H > 0.5$: persistent memory) via $H \approx 1 - \beta/2$ for stretched exponential processes and $H = (1 + \alpha)/2$ for $1/f^{\alpha}$ noise.

A known objection to universality claims is that any ensemble of many independent exponentially relaxing modes with a broad rate distribution generically produces $H \approx 0.7$–0.8 [31]. This superposition argument applies to bulk/aggregate measurements (glasses, financial markets, river flows, EEG) where many components are averaged. It does *not* apply to single-molecule and single-system measurements, where there is no ensemble to superpose. But the single-entity restriction excludes only the *superposition* explanation: non-exponential single-molecule kinetics still arises from coupling to slow internal degrees of freedom — and classical dynamic disorder (the fluctuating-enzyme picture; Zwanzig 1990; Lu et al. 1998) is exactly such a single-entity, intrinsic mechanism, one the framework's own §7.4 notes is model-independent. This is the C1–C4 architecture in the weak (grade-ii, memory) sense of the §7.2 scope note; it is consistent with — but does not isolate — the framework's distinctive P-indivisibility.

Restricting to single-entity measurements where superposition is excluded by construction:

System	Measurement	Memory depth	H	Source
β-galactosidase	Single-molecule turnover	$\sim 10^{2.6}$	0.80	English et al. 2006 ($\beta = 0.4$)
Cholesterol oxidase	Single-molecule turnover	$\sim 10^{1.6}$	0.75	Lu et al. 1998 ($\beta \approx 0.5$)
Horseradish peroxidase	Single-molecule turnover	$\sim 10^{3}$	0.83	Edman/Rigler 2000 ($\beta \approx 0.35$)
Lipase B	Single-molecule turnover	$\sim 10^{2}$	0.80	Flomenbom et al. 2005 ($\beta \approx 0.4$)
Myoglobin CO rebinding	Single-molecule kinetics	$\sim 10^{6}$	0.95	Austin et al. 1975 ($\beta \approx 0.1$)

System	Measurement	Memory depth	H	Source
SC qubit (Lorentzian TLS)	Single-qubit trajectory	$\sim 10^2$	0.80	Noise spectroscopy ($\alpha \approx 0.6$)
Ion channel (membrane-coupled)	Single-channel recording	$\sim 10^3$	0.80	Dwell-time autocorrelation

For the four enzymes and the qubit — all with "typical" C3 capacity (conformational state spaces of $\sim 10^2$–10^4 states) — $H = 0.79 \pm 0.03$. Two qualifications keep this honest. First, H here is not an independent measurement: it is the measured stretched/1-f exponent rescaled by the conversions stated above ($H = 1 - \beta/2$, $H = (1 + \alpha)/2$), so "universal $H = 0.79 \pm 0.03$" is equivalent to "these selected systems have $\beta \approx 0.42$," a known range for single-molecule dispersed kinetics — the regularity is inherited from the selection, not discovered across it. Second, the superposition objection is excluded, but the classical dynamic-disorder mechanism is not: the memory is intrinsic to each single entity, and arises from coupling to a slow internal hidden sector (protein scaffold conformational dynamics for enzymes, TLS bath for the qubit), which is a grade-ii signature reproduced by classical (divisible) fluctuating-rate models. It is therefore a consistency observation, not a distinguishing test of P-indivisibility.

The myoglobin outlier ($H = 0.95$) has an anomalously deep conformational hierarchy — Frauenfelder's "protein energy landscape" involves $\sim 10^6$–10^9 conformational substates organized in a fractal-like tier structure, far exceeding typical enzyme C3 capacity. This suggests H increases with C3: typical capacity gives $H \approx 0.8$; deep capacity gives $H \to 1$. The genuinely distinctive content of this section is not the single-system value (degenerate, above) but this *cross-substrate* law $H(C3)$ — the claim that one memory-strength relation holds across an enzyme, a qubit, and an ion channel, which per-system classical models do not unify. As stated, however, it is derived only "in principle" from the characterization theorem's dependence on hidden-sector dimension, and it is fit in-sample (the single deep point, myoglobin, motivates it). The operative test is therefore out-of-sample: measure H on systems of independently known hidden-sector dimension and check the predicted $H(C3)$.

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https://github.com/amaybaum/incompleteness#ai-assistance.

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